SQL Code

```sql
show databases;

create database RetailDB_2 ;

use RetailDB_2 ;

show tables;

CREATE TABLE Customers (
customer_id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100),
city VARCHAR(50),
signup_date DATE
);

insert into Customers(customer_id, name, email, city, signup_date)values
(1,"Rohit Kumar",  "rohit.kumar@gmail.com",   "Delhi",      '2023-01-12'),
(2,"Sneha Sharma", "sneha.sharma@yahoo.com",  "Mumbai",     '2023-02-05'),
(3,"Amit Patel",   "amit.patel@gmail.com",    "Ahmedabad",      '2023-02-18'),
(4,"Priya Reddy",  "priya.reddy@gmail.com",   "Hyderabad",      '2023-03-02'),
(5,"Karan Singh",  "karan.singh@outlook.com", "Chennai",    '2023-03-15'),
(6,"Neha Verma",   "neha.verma@gmail.com",    "Pune",       '2023-04-01'),
(7,"Arjun Mehta",  "arjun.mehta@gmail.com",   "Bengaluru",      '2023-04-20'),
(8,"Ritika Gupta", "ritika.gupta@yahoo.com",  "Kolkata",        '2023-05-12'),
(9,"Vikram Joshi", "vikram.joshi@gmail.com",  "Lucknow",        '2023-05-25'),
(10,"Ananya Das",  "ananya.das@gmail.com",        "Bhubaneswar",'2023-06-08'),
(11,"Suresh Iyer", "suresh.iyer@gmail.com",   "Chennai",        '2023-06-20'),
(12,"Megha Kapoor","megha.kapoor@yahoo.com",  "Jaipur",     '2023-07-03'),
(13,"Ravi Shankar","ravi.shankar@gmail.com",  "Delhi",      '2023-07-15'),
(14,"Tanya Mishra","tanya.mishra@gmail.com",  "Noida",      '2023-08-01'),
(15,"Aditya Jain", "aditya.jain@gmail.com",       "Indore",     '2023-08-14');

select * from  Customers;

CREATE TABLE Orders (
order_id INT AUTO_INCREMENT PRIMARY KEY,
customer_id INT,
order_date DATE,
total_amount DECIMAL(10,2),
payment_mode VARCHAR(50),
FOREIGN KEY (customer_id) REFERENCES Customers(customer_id));

insert into orders(order_id,    customer_id,    order_date,     total_amount,
payment_mode)values
(1      ,1      ,"2024-01-05"   ,"79999", "UPI"),
(2      ,2      ,"2024-01-08"   ,"74999"        ,"Credit Card"),
(3      ,3      ,"2024-01-10"   ,"9998" ,"Debit Card"),
(4      ,4      ,"2024-01-15"   ,"2499" ,"UPI"),
(5      ,5      ,"2024-01-18"   ,"5997" ,"Cash"),
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```
(6      ,6      ,"2024-01-20"   ,"2999" ,"UPI"),
(7      ,7      ,"2024-01-22"   ,"6499" ,"Credit Card"),
(8      ,8      ,"2024-01-25"   ,"59999"        ,"Net Banking"),
(9      ,9      ,"2024-01-28"   ,"55999"        ,"UPI"),
(10     ,10     ,"2024-01-30"   ,"2998" ,"Debit Card"),
(11     ,11     ,"2024-02-02"   ,"79999"        ,"Credit Card"),
(12     ,12     ,"2024-02-05"   ,"74999"        ,"Net Banking"),
(13     ,13     ,"2024-02-08"   ,"4999" ,"UPI"),
(14     ,14     ,"2024-02-10"   ,"4998" ,"Cash"),
(15     ,15     ,"2024-02-12"   ,"1999" ,"UPI"),
(16     ,6      ,"2024-02-14"   ,"5998" ,"Credit Card"),
(17     ,7      ,"2024-02-16"   ,"6499" ,"UPI"),
(18     ,8      ,"2024-02-18"   ,"59999"        ,"Debit Card"),
(19     ,9      ,"2024-02-20"   ,"111998"       ,"Credit Card"),
(20     ,10     ,"2024-02-22"   ,"1499" ,"Cash"),
(21     ,11     ,"2024-02-24"   ,"3998" ,"UPI"),
(22     ,12     ,"2024-02-26"   ,"4999" ,"Net Banking"),
(23     ,13     ,"2024-02-28"   ,"7497" ,"Credit Card"),
(24     ,14     ,"2024-03-01"   ,"2999" ,"UPI"),
(25     ,15     ,"2024-03-03"   ,"79999"        ,"Debit Card"),
(26     ,2      ,"2024-03-05"   ,"6499" ,"Credit Card"),
(27     ,5      ,"2024-03-07"   ,"59999"        ,"UPI"),
(28     ,9      ,"2024-03-10"   ,"74999"        ,"Cash"),
(29     ,13     ,"2024-03-12"   ,"55999"        ,"Net Banking"),
(30     ,14     ,"2024-03-14"   ,"2998" ,"UPI"),
(31     ,1      ,"2024-03-16",  "1499", "Cash"),
(32     ,3      ,"2024-03-18",  "2999", "UPI"),
(33     ,4      ,"2024-03-20",  "1999", "Debit Card"),
(34     ,5      ,"2024-03-22",  "4999", "Credit Card"),
(35     ,6      ,"2024-03-24",  "74999",        "Net Banking"),
(36     ,7      ,"2024-03-26",  "55999",        "Cash"),
(37     ,8      ,"2024-03-28",  "79999",        "UPI"),
(38     ,9      ,"2024-03-30",  "2999", "Credit Card"),
(39     ,10     ,"2024-04-01",  "6499", "UPI"),
(40     ,11     ,"2024-04-03",  "55999",        "Debit Card"),
(41     ,12     ,"2024-04-05",  "1999", "Cash"),
(42     ,13     ,"2024-04-07",  "2499", "Net Banking"),
(43     ,14     ,"2024-04-09",  "2998", "UPI"),
(44     ,15     ,"2024-04-11",  "79999",        "Credit Card"),
(45     ,2      ,"2024-04-13",  "74999",        "Net Banking"),
(46     ,3      ,"2024-04-15",  "5999", "UPI"),
(47     ,5      ,"2024-04-17",  "7999", "Credit Card"),
(48     ,7      ,"2024-04-19",  "2999", "Cash"),
(49     ,9      ,"2024-04-21",  "9999", "Debit Card"),
(50     ,8      ,"2024-04-23",  "59999",        "Net Banking");

select * from orders;

CREATE TABLE Suppliers (
supplier_id INT AUTO_INCREMENT PRIMARY KEY,
supplier_name VARCHAR(100),
contact_email VARCHAR(100),
city VARCHAR(50));
```

```sql
insert into Suppliers (supplier_id, supplier_name, contact_email, city)values
(1,     "Reliance Digital",     "support@reliance.com", "Mumbai"),
(2,     "Croma Electronics",    "info@croma.com"        ,"Delhi"),
(3,     "Flipkart Seller Hub",  "sellers@flipkart.com", "Bengaluru"),
(4,     "Amazon India Vendor",  "vendor@amazon.in"      ,"Hyderabad"),
(5,     "Tata Cliq"     ,"support@tatacliq.com",        "Pune"),
(6,     "Vijay Sales"   ,"sales@vijaysales.com",        "Chennai"),
(7,     "Lifestyle Stores"      ,"info@lifestylestores.com",    "Kolkata"),
(8,     "Metro Electronics",    "metro@electronics.com",        "Ahmedabad");

select * from Suppliers;

CREATE TABLE Products (
product_id INT AUTO_INCREMENT PRIMARY KEY,
product_name VARCHAR(100),
category VARCHAR(50),
price DECIMAL(10,2),
stock_qty INT,
supplier_id INT,
FOREIGN KEY (supplier_id) REFERENCES Suppliers(supplier_id)
);

insert into Products(product_id, product_name, category, price, stock_qty,supplier_id)values
(1      ,"iPhone 15"                ,"Electronics"          ,"79999"      ,'50'   ,'1'),
(2      ,"Samsung Galaxy S24"       ,"Electronics"          ,"74999"      ,'40'   ,'2'),
(3      ,"Noise Smartwatch"         ,"Wearables"            ,"4999"       ,'100' ,'3'),
(4      ,"Boat Earbuds"             ,"Wearables"            ,"2499"       ,'200' ,'4'),
(5      ,"Kurta Set"                ,"Fashion"          ,"1999"   ,'150' ,'7'),
(6      ,"Running Shoes"            ,"Fashion"          ,"2999"   ,'120' ,'7'),
(7      ,"Prestige Mixer Grinder"   ,"Home Appliances" ,"6499"   ,'80'   ,'5')   ,
(8      ,"Sony Bravia TV"               ,"Electronics"      ,"59999"      ,'30'   ,'6'),
(9      ,"Lenovo Laptop"            ,"Electronics"          ,"55999"      ,'45'   ,'2'),
(10     ,"Philips Trimmer"          ,"Personal Care"        ,"1499"       ,'90'   ,'8');

select * from Products;

CREATE TABLE Order_Items (
order_item_id INT AUTO_INCREMENT PRIMARY KEY,
order_id INT,
product_id INT,
quantity INT,
price_each DECIMAL(10,2),
FOREIGN KEY (order_id) REFERENCES Orders(order_id),
FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

insert into Order_Items(order_item_id, order_id, product_id, quantity, price_each)values
(1      ,"1"    ,"1"    ,"1"    ,"79999"),
(2      ,"2"    ,"2"    ,"1"    ,"74999"),
```

```
(3      ,"3"    ,"3"    ,"2"    ,"4999"),
(4      ,"4"    ,"4"    ,"1"    ,"2499"),
(5      ,"5"    ,"5"    ,"3"    ,"1999"),
(6      ,"6"    ,"6"    ,"1"    ,"2999"),
(7      ,"7"    ,"7"    ,"1"    ,"6499"),
(8      ,"8"    ,"8"    ,"1"    ,"59999"),
(9      ,"9"    ,"9"    ,"1"    ,"55999"),
(10     ,"10"   ,"10"   ,"2"    ,"1499"),
(11     ,"11"   ,"1"    ,"1"    ,"79999"),
(12     ,"12"   ,"2"    ,"1"    ,"74999"),
(13     ,"13"   ,"3"    ,"1"    ,"4999"),
(14     ,"14"   ,"4"    ,"2"    ,"2499"),
(15     ,"15"   ,"5"    ,"1"    ,"1999"),
(16     ,"16"   ,"6"    ,"2"    ,"2999"),
(17     ,"17"   ,"7"    ,"1"    ,"6499"),
(18     ,"18"   ,"8"    ,"1"    ,"59999"),
(19     ,"19"   ,"9"    ,"2"    ,"55999"),
(20     ,"20"   ,"10"   ,"1"    ,"1499"),
(21     ,"21"   ,"5"    ,"2"    ,"1999"),
(22     ,"22"   ,"3"    ,"1"    ,"4999"),
(23     ,"23"   ,"4"    ,"3"    ,"2499"),
(24     ,"24"   ,"6"    ,"1"    ,"2999"),
(25     ,"25"   ,"1"    ,"1"    ,"79999"),
(26     ,"26"   ,"7"    ,"1"    ,"6499"),
(27     ,"27"   ,"8"    ,"1"    ,"59999"),
(28     ,"28"   ,"2"    ,"1"    ,"74999"),
(29     ,"29"   ,"9"    ,"1"    ,"55999"),
(30     ,"30"   ,"10"   ,"2"    ,"1499"),
(31     ,"31"   ,"6"    ,"1"    ,"2999"),
(32     ,"32"   ,"5"    ,"1"    ,"1999"),
(33     ,"33"   ,"3"    ,"1"    ,"4999"),
(34     ,"34"   ,"2"    ,"1"    ,"74999"),
(35     ,"35"   ,"9"    ,"1"    ,"55999"),
(36     ,"36"   ,"1"    ,"1"    ,"79999"),
(37     ,"37"   ,"7"    ,"1"    ,"6499"),
(38     ,"38"   ,"6"    ,"1"    ,"2999"),
(39     ,"39"   ,"9"    ,"1"    ,"55999"),
(40     ,"40"   ,"2"    ,"10"   ,"74999"),
(41     ,"41"   ,"5"    ,"1"    ,"1999"),
(42     ,"42"   ,"4"    ,"1"    ,"2499"),
(43     ,"43"   ,"10"   ,"2"    ,"1499"),
(44     ,"44"   ,"1"    ,"1"    ,"79999"),
(45     ,"45"   ,"2"    ,"1"    ,"74999"),
(46     ,"46"   ,"3"    ,"1"    ,"5999"),
(47     ,"47"   ,"5"    ,"2"    ,"1999"),
(48     ,"48"   ,"6"    ,"1"    ,"2999"),
(49     ,"49"   ,"9"    ,"1"    ,"9999"),
(50     ,"50"   ,"8"    ,"1"    ,"59999");
 select * from Order_Items;


 ### a) Normal Queries
### Q1. Fetch all products along with their supplier name (INNER JOIN).

```

```
select * from Products;
select * from Suppliers;
select product_name,supplier_name from Products P inner join Suppliers S on P.supplier_id=S.supplier_id;


### Q2. Find all customers and their orders, even if they have not placed any (LEFT JOIN).

select *from Customers;
select *from Orders;
select *from Customers C left join Orders O on C.customer_id=O.customer_id;


### Q3. Get all suppliers and the products they supply, even if no products exist for a supplier (RIGHT JOIN).

select supplier_name, product_name from Products P right join Suppliers S on P.supplier_id=S.supplier_id;


### Q4. Show all customers and all orders (FULL OUTER JOIN simulation using UNION).

select *from Customers C join Orders O on C.customer_id=O.customer_id;


### Q5. List all products priced between ₹5000 and ₹50,000 and supplied from "Mumbai".

select * from Products;
select * from Suppliers;
select product_name,price,city from Products P  join Suppliers S on P.supplier_id=S.supplier_id where city='Mumbai' and  price between 5000 and 50000;

  ###b)  Aggregations & Group By

###Q6. Find the total number of orders placed by each customer and show only those who placed more than 2 (GROUP BY + HAVING).

select *from Orders;
select customer_id, count(customer_id) T_O from Orders group by customer_id having count(customer_id)>2 order by customer_id;


###Q7. Show each supplier’s total sales value (sum of quantity × price_each).

select *from Products;
select supplier_name, price*stock_qty as total_sale from Products P join Suppliers S on P.supplier_id=S.supplier_id;


###Q8. Find the average, highest, and lowest price of products in each category.

select *from Products;
select category,count(category) NO_OF_ITEM from Products group by category ;
```

```
select category,count(category) NO_OF_ITEM, avg(price) Average, max(price) Highest, min(price) Lowest from Products where category='Electronics'group by category union
select category,count(category) NO_OF_ITEM, avg(price) Average, max(price) Highest, min(price) Lowest from Products where category='Wearables'group by category union
select category,count(category) NO_OF_ITEM, avg(price) Average, max(price) Highest, min(price) Lowest from Products where category='Fashion'group by category union
select category,count(category) NO_OF_ITEM, avg(price) Average, max(price) Highest, min(price) Lowest from Products where category='Home Appliances'group by category union
select category,count(category) NO_OF_ITEM, avg(price) Average, max(price) Highest, min(price) Lowest from Products where category='Personal Care'group by category ;


###Q9. Find the top 5 customers by total spending (ORDER BY SUM(total_amount) DESC LIMIT 5).

select * from orders;
select customer_id , sum(total_amount) total_spends from Orders group by customer_id order by total_spends desc limit 5;


###Q10. Show the number of unique products ordered by each customer.

select name,count(name) no_of_product from
(select name,product_name from order_items OT join products P on P.product_id=OT.product_id join orders O on O.order_id=OT.order_id
join customers C on C.customer_id=O.customer_id group by name,product_name) a group by name;


 ###c)  Subqueries

### Q11. Find customers who placed an order with an amount greater than the average order amount (subquery).

select avg(total_amount) from orders;
select customer_id, total_amount from orders where total_amount > (select avg(total_amount) from orders) order by total_amount;


### Q12. Find products that have never been ordered (subquery with NOT IN).

select product_name from products where product_name not in
(select product_name from products P join orders O on P.price=O.total_amount group by product_name);


### Q13. List customers who ordered at least one product from the "Electronics" category.

 select * from orders;
 select customer_id,category from products P join orders O on P.price=O.total_amount where category='electronics' group by customer_id;

```

```

### Q14. Get suppliers who provide products that have been ordered more than 100 times in total.

select stock_qty,supplier_id from products where stock_qty>100;

### Q15. Find the most expensive product(s) using a subquery with MAX().
select max(price) as max_price from products;
select product_name,price  from products where price =(select max(price) from products);


### d)  Advanced Filters

### Q16. Show orders placed by customers who live in either Mumbai, Delhi, or Bengaluru (IN operator).

select name,order_id,city from orders O join customers C on O.customer_id=C.customer_id where city in ('Mumbai','Delhi','Bengaluru');


### Q17. Show orders where payment mode is NOT UPI or Credit Card (NOT IN).

select * from orders where payment_mode not in ("UPI","Credit Card");


### Q18. Find customers who have no email address recorded (IS NULL).

select * from customers where email=null;

### Q19. Show suppliers who are not from the same city as any customer (NOT IN subquery).


 select *from suppliers S join customers C on S.city=C.city where S.city not in(select city from customers);


### Q20. Get the latest 3 orders placed, skipping the first 2 (ORDER BY + LIMIT + OFFSET).

select * from orders;

select * from orders order by order_date asc limit 3 ,2;
```